Lecture 8 - Causal Inference with Panel Data

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Revisit Causal Model

Model-based analysis (for non-experimental data)

- Recall OLS gives the best linear approximation to $E[y_i|x_i]$. But in many situations we are not really interested in approximating $E[y_i|x_i]$. Instead, we are are interested in a *"causal model"* that describes the determinants of y:

$$y_i = x_i'\beta + \nu_i$$

  in which:

  1. we know that y depends on observed $x's$ and unobserved stuff ν
  2. we are interested in $\partial y / \partial x_k$ holding constant ν
  3. we are concerned that some of the things in ν are correlated with x

Revisit Causal Model

- Notice that the *"causal model"*:

$$y_i = x_i'\beta + \nu_i$$

  Looks just like a regression model, which is very confusing.

- When people say they are writing a causal model, what they are doing is giving the true "data generating process": (the "true DGP") that specifies how outcome y_i depends on <u>observed</u> inputs x_i (through the function $x_i'\beta$) and <u>unobserved</u> inputs ν_i

- If the causal model is:

$$y_i = x_i'\beta + \nu_i$$

  and $E[x_i\nu_i] \neq 0$, then the population regression is:

$$\begin{aligned} \beta^{OLS} &= \beta + \pi \\ \pi &= E[x_i x_i']^{-1} E[x_i \nu_i] \end{aligned}$$

  i.e., π is the vector of regression coefficients when we regress ν_i on x_i (which we can't really do)!

Revisit Causal Model

- Model based approach: make *assumptions* on the nature of ν_i, and how its related to x_i. Then develop an approach to remove the "bad" part of ν_i
- Today, we are gonna apply model-based approach to panel data, where we observe the same unit for multiple time periods.

This Lecture

- Introduce panel data: what are panel data and why are panel data useful?
- Examples:
  - Workers observed in 2 or more periods
  - Test scores of kids observed in different grades
  - Employment of fast food restaurants observed in different years
- This lecture will be the building block for understanding difference-in-differences & event study...

What are panel data?

- Panel data has a group structure: units of observation can be "grouped"
  - y_{it}: wage of person i in period t, the data is at person$\times$time period level. Group = person.
  - R_{ic}: rent paid by family i in city c. One may be interested in group = city. What is a city's fixed effect in rents?
  - We will look **within** and **between** groups.

Why are panel data useful?

- Suppose the "true" model generating wages is:

$$y_i = x_i' \beta + \nu_i$$

  where x includes the usual variables (age, education, gender) and a dummy for government sector job. This is *not* the way we have been describing regression models up to now. Instead, this is a model-based approach where we say that the "true" coefficients of interest are β.

- What happens when we run a regression? Omitted variable formula says:

$$\beta^o = \beta + \pi$$

  where π comes from the auxiliary regression: $\nu_i = x_i' \pi + \epsilon_i$

$$\pi = E[x_i x_i']^{-1} E[x_i \nu_i]$$

- Assume $x_i' = (z_i', D_i)$ where z is a $K - 1$ vector that includes a constant and other controls.

Why are panel data useful?

- Apply Frisch-Waugh to the auxiliary regression $\nu_i = z_i'\pi_{1\ldots(K-1)} + \pi_K D_i + \epsilon_i$, we know the last element of the coefficient vector π is

$$\pi_K = E[\xi_i^2]^{-1} E[\xi_i \nu_i]$$

where ξ_i is the residual from a population regression of D_i on all the other elements of x_i (i.e. z_i) :

$$D_i = z_i'\tau + \xi_i.$$

So if D_i is not related to the other $z's$, π_K is just the *difference in all unobserved factors that determine wages* between the government and private sector. Otherwise its the difference, after taking out whatever can be explained by the difference in $z's$ across sectors.

Why are panel data useful?

- To recap: we assume a "true model"

$$y_i = x_i'\beta + \nu_i$$

We have $x_i' = (z_i', D_i)$, and we're interested in the coefficient on the dummy D_i:

$$\beta_K^o = \beta_K + \pi_K = \beta_K + E[\xi_i^2]^{-1}E[\xi_i\nu_i]$$

where ξ_i is the part of D_i we can't explain from the $z's$. The 2nd term ("bias") can be interpreted as the "difference in unobserved ability between the public and the private sector".

Why are panel data useful?

- If we are interested in trying to get a "pure" estimate of β_K – the "wage effect" of working in the government sector, after taking account of ability differences – we need another approach.
- Suppose we see the same worker in 2 periods (panel data!), and sometimes she is working in one sector, and sometimes in the other (mover design). Write the model for periods 1 and 2 as:

$$
\begin{aligned}
y_{i1} &= x_{i1}'\beta + \nu_{i1} \\
y_{i2} &= x_{i2}'\beta + \nu_{i2}
\end{aligned}
$$

- Now let's make the *modeling assumption* that:

$$\nu_{it} = \alpha_i + \epsilon_{it}$$

  So we are breaking out the unobserved stuff into two parts: a permanent component α_i, and time-varying components ϵ_{it}.
- Now the model can be written:

$$
\begin{aligned}
y_{i1} &= x_{i1}'\beta + \alpha_i + \epsilon_{i1} \\
y_{i2} &= x_{i2}'\beta + \alpha_i + \epsilon_{i2}
\end{aligned}
$$

Why are panel data useful?

- Suppose that we think all the “problems” that cause a correlation of (x_{it}, ν_{it}) arise because of the permanent stuff. So $E[x_{it}\epsilon_{it}] = 0$. Then we can take differences:

$$\triangle y_i = y_{i2} - y_{i1} = \triangle x_i \beta_1 + \triangle \epsilon_i$$

  and **“difference out”** the ability term!
- The key advantage of panel data is that we can potentially eliminate the effect of any variables that are constant within the “group”. To see this more generally, write the model as

$$y_{it} = x_{it}'\beta + \alpha_i + \epsilon_{it} \qquad t = 1, 2, \ldots T;\ i = 1, 2..N \qquad (*)$$

  Then consider the averages: $\overline{y}_i, \overline{x}_i, \overline{\epsilon}_i$ (e.g., $\overline{y}_i = \frac{1}{T}\sum_{t=1}^{T} y_{it}$). Clearly:

$$\overline{y}_i = \overline{x}_i\beta + \alpha_i + \overline{\epsilon}_i \qquad (**)$$

  (You should check this by taking the average of both sides of (*).

Within estimator

- Subtracting the time averages from each period we get:

$$y_{it} - \overline{y}_i = (x_{it} - \overline{x}_i)'\beta + \epsilon_{it} - \overline{\epsilon}_i \qquad (***)$$

- This says that in a panel data setting, we can get the coefficients for the time-varying $x's$ by deviating the data from group means! Note that if there is some element in x_{it} that does not vary within the group, it "drops out", along with the α_i term. (This means we don't get an estimate of the constant coefficient).
- Deviating the (y_{it}, x_{it}) data for each observation in a group from the group means $(\overline{y}_i, \overline{x}_i)$ is called a **"within-group" transformation**. The OLS estimator of β after deviating from group means is called the **"within" estimator.**

Within estimator

- If you stare at $(***)$ and think about it, you will see that although there are T deviations from means for each i, they always sum to 0: $\sum_{t=1}^{T}(y_{it} - \bar{y}_i) = 0$. So you can drop one of the observations for each group.
- In the case where we have 2 observations per group

$$y_{i2} - \overline{y}_i = y_{i2} - \frac{y_{i1} + y_{i2}}{2} = \frac{y_{i2} - y_{i1}}{2}$$

and similarly for each row of x_{it}, so with only 2 observations per group, deviating from means is the same as **differencing / first-difference** model.

OLS estimators: "Cross-sectional", Within, Between

- You could also get an estimate of β from the group means equation $(**)$: $\overline{y}_i = \overline{x}_i\beta + \alpha_i + \overline{\epsilon}_i$. This estimate is called the "between" estimator. The estimation assumes away
- The between and within estimators will not necessarily be equal. It is sometimes interesting to compare the three possible OLS estimators:

$$
\begin{aligned}
\hat{\beta} &= \left(\frac{1}{NT}\sum_{i,t}(x_{it}x_{it}')\right)^{-1}\frac{1}{NT}\sum_{i,t}x_{it}y_{it} \qquad (1)\\
\hat{\beta}^B &= \left(\frac{1}{N}\sum_{i}(\overline{x}_i\overline{x}_i')\right)^{-1}\frac{1}{N}\sum_{i}\overline{x}_i\overline{y}_i\\
\hat{\beta}^W &= \left(\frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(x_{it}-\overline{x}_i)'\right)^{-1}\frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i)
\end{aligned}
$$

  - If we are more concerned about between-group differences, the between estimator is more relevant – it picks up the effect of any fixed unobserved effects in each group most strongly.

OLS estimators: "Cross-sectional", Within, Between

- It can be shown that $\hat{\beta}$ is a weighted average of $\hat{\beta}^B$ and $\hat{\beta}^W$:

$$\frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(x_{it}-\overline{x}_i)' = \frac{1}{NT}\sum_i\sum_t x_{it}x_{it}' - 2\bar{x}_i x_{it}' + \overline{x}_i\bar{x}_i'$$

$$= \frac{1}{NT}\sum_{i,t} x_{it}x_{it}' - \frac{1}{N}\sum_i \bar{x}_i\bar{x}_i'$$

$$\Rightarrow \frac{1}{NT}\sum_{i,t} x_{it}x_{it}' = \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(x_{it}-\overline{x}_i)' + \frac{1}{N}\sum_i(\overline{x}_i\overline{x}_i')$$

$$\frac{1}{NT}\sum_{i,t} x_{it}y_{it} = \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i) + \frac{1}{N}\sum_i \bar{x}_i\bar{y}_i$$

- Define $\widetilde{\Omega}_{xy} = \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i)$, $\widetilde{\Omega}_{xx} = \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(x_{it}-\overline{x}_i)'$. Then the within estimator $\hat{\beta}^W = (\widetilde{\Omega}_{xx})^{-1}\widetilde{\Omega}_{xy}$. Define $\overline{\Omega}_{xy} = \frac{1}{N}\sum_i \bar{x}_i\bar{y}_i$, and $\overline{\Omega}_{xx} = \frac{1}{N}\sum_i \bar{x}_i\bar{x}_i'$. Then the between estimator $\hat{\beta}^B = (\overline{\Omega}_{xx})^{-1}\overline{\Omega}_x$.

OLS estimators: "Cross-sectional", Within, Between

- Express $\hat{\beta}$ as a combination of within and between estimators:

$$\hat{\beta} = \left(\frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i)(x_{it} - \overline{x}_i)' + \frac{1}{N} \sum_i (\overline{x}_i \overline{x}_i') \right)^{-1} * \left(\frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i)(y_{it} - \overline{y}_i) + \frac{1}{N} \sum_i \bar{x}_i \bar{y}_i \right) \tag{2}$$

$$= \Omega_{xx}^{-1} \left(\widetilde{\Omega}_{xy} + \overline{\Omega}_{xy} \right)$$

$$= \Omega_{xx}^{-1} \underbrace{\widetilde{\Omega}_{xx} \left(\widetilde{\Omega}_{xx} \right)^{-1}}_{I_K} \left(\widetilde{\Omega}_{xy} \right) + \Omega_{xx}^{-1} \underbrace{\overline{\Omega}_{xx} \left(\overline{\Omega}_{xx} \right)^{-1}}_{I_K} \left(\overline{\Omega}_{xy} \right)$$

$$= \Omega_{xx}^{-1} \widetilde{\Omega}_{xx} \underbrace{\left(\widetilde{\Omega}_{xx} \right)^{-1} \left(\widetilde{\Omega}_{xy} \right)}_{\text{within}} + \Omega_{xx}^{-1} \overline{\Omega}_{xx} \underbrace{\left(\overline{\Omega}_{xx} \right)^{-1} \left(\overline{\Omega}_{xy} \right)}_{\text{between}}$$

$$= \Omega_{xx}^{-1} \widetilde{\Omega}_{xx} \hat{\beta}^W + \Omega_{xx}^{-1} \overline{\Omega}_{xy} \hat{\beta}^B \tag{3}$$

OLS estimators: "Cross-sectional", Within, Between

- (Extension) Related to the Law of Total Variance:

$$var(Y) = E[var(Y|X)] + var(E[Y|X]).$$

Proof: Define $\mu_Y = E[Y]$,and $\mu_{Y|X} = E[Y|X]$. Apply the law of iterated expectations:

$$\begin{aligned}
E[(Y-\mu_Y)^2] =& E_X[E[(Y-\mu_Y)^2|X]] \\
=& E_X[E[(Y-\mu_{Y|X}+\mu_{Y|X}-\mu_Y)^2|X]] \\
=& E_X[E[(Y-\mu_{Y|X})^2|X] + E_X[\underbrace{(\mu_{Y|X}-\mu_Y)^2}_{\text{constant } | \text{ X}}] \\
& + 2E_X[\underbrace{E[(Y-\mu_{Y|X})|X]}_{=0} \times \underbrace{(\mu_{Y|X}-\mu_Y)}_{\text{constant } | \text{ X}}] \\
=& E_X[var(Y|X)] + var(E[Y|X]) = \text{within} + \text{between}
\end{aligned}$$

In equation (2), we decompose the (total) covariance of X and Y into a within component and a between component.

Summary - OLS Estimators

Let's look at the 3 estimating equations:

$$\begin{aligned} y_{it} &= x_{it}'\beta + \alpha_i + \epsilon_{it} \\ \overline{y}_i &= \overline{x}_i\beta + \alpha_i + \overline{\epsilon}_i \\ y_{it} - \overline{y}_i &= (x_{it} - \overline{x}_i)'\beta + \epsilon_{it} - \overline{\epsilon}_i \end{aligned} \tag{4}$$

in each case we express the relationship between an outcome and a set of observed and unobserved components, with some assumptions. Our key assumptions are:

$$\begin{aligned} E[x_{it}\alpha_i] \neq 0 \\ E[x_{it}\epsilon_{it}] = 0 \end{aligned} \tag{5}$$

the correlation of the residual with x_{it} is confined to α_i. We further assume the correlation between x_{it} and α_i is time-invariant: $E[(x_{it} - \bar{x}_i)\alpha_i] = 0$.

Summary - OLS Estimators

Suppose that $x'_{it} = (1, z_{it})$, where z_{it} is a scalar. Then applying the usual FW logic to regressions with a constant and a single covariate:

$$\hat{\beta}_2 = \left(\frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z})^2 \right)^{-1} \frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z})(z_{it}\beta_2 + \alpha_i + \epsilon_{it})$$

$$\hat{\beta}_2^B = \left(\frac{1}{N} \sum_{i} (\bar{z}_i - \bar{z})^2 \right)^{-1} \frac{1}{N} \sum_{i} (\bar{z}_i - \bar{z})(\bar{z}_i\beta_2 + \alpha_i + \bar{\epsilon}_i)$$

$$\hat{\beta}_2^W = \left(\frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z})^2 \right)^{-1} \frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z}_i)\left((z_{it} - \bar{z}_i)\beta_2 + \epsilon_{it} - \bar{\epsilon}_i\right)$$

in which $\bar{z} = \frac{1}{N} \sum_i \bar{z}_i = \frac{1}{NT} \sum_{i,t} z_{it}$.

Summary - OLS Estimators

Let's look further,

$$\hat{\beta}_2 = \beta_2 + \left(\frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z})^2\right)^{-1} \underbrace{\frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z})\alpha_i}_{(*)}$$

$$+ \left(\frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z})^2\right)^{-1} \underbrace{\frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z})\epsilon_{it}}_{\xrightarrow{p} 0}$$

the "bias" term in OLS $\hat{\beta}_2$ comes from $(*)$:

$$\begin{aligned} \frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z})\alpha_i &= \frac{1}{NT}\sum_{i,t}(z_{it}-\bar{z}_i+\bar{z}_i-\bar{z})\alpha_i \\ &= \frac{1}{N}\sum_{i,t}(\bar{z}_i-\bar{z})\alpha_i \end{aligned}$$

in which $z_{it}-\bar{z}_i \perp \alpha_i$ as we assume the correlation between observable z_{it} and α_i is time-invariant.

Summary - OLS Estimators

Comparing that with between-estimator $\hat{\beta}_2^B$:

$$\hat{\beta}_2^B = \beta_2 + \left(\frac{1}{N} \sum_i (\bar{z}_i - \bar{z})^2 \right)^{-1} \frac{1}{N} \sum_i (\bar{z}_i - \bar{z})(\alpha_i + \bar{\epsilon}_i)$$

$$\text{"bias"} = \left(\frac{1}{N} \sum_i (\bar{z}_i - \bar{z})^2 \right)^{-1} \frac{1}{N} \sum_i (\bar{z}_i - \bar{z})\alpha_i$$

which has the same term $\frac{1}{N} \sum_i (\bar{z}_i - \bar{z})\alpha_i$, but a different weight matrix (instead of $\left(\frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z})^2 \right)^{-1}$).

A closer look at the within transformation

- It is painful to construct the "deviations from group means" for all the $x's$ and for y. So people don't normally construct $\hat{\beta}^W$ by hand. Instead they rely on the following fact.
- Consider the expanded model:

$$y_{it} = x'_{it}\beta + (D_{1i}, D_{2i}, ...D_{Ni})\theta + \alpha_i + \epsilon_{it} \tag{6}$$

  where D_{gi} is a dummy variable for "person g" ($D_{gi} = 1[g = i]$). This is a model in which we add N dummies, one for each person (and take the constant out of x). Then the OLS estimator from this model is $\hat{\beta}^W$. Why? Frisch Waugh!
- Using FW, we know that you get the same coefficients for the $x's$ if you first regress the $x's$ on the dummies, then take the residuals, and regress y on the residuals. Let's think what this does...

A closer look at the within transformation

- First we regress x_{kit} on the person dummies for each k. But we know from Lecture 9 that the coefficents for the dummies will just be the person means of x_{kit}. In other words, if we fit the model:

$$x_{kit} = (D_{1i}, D_{2i}, ...D_{Ni}) \begin{pmatrix} \theta_1^k \\ ... \\ \theta_N^k \end{pmatrix} + \xi_{kit} \tag{7}$$

we'll get that the i^{th} row of $\hat{\theta}^k$ will be $\overline{x}_{ki}$. So the "residual" will be $\xi_{kit} = x_{kit} - \overline{x}_{ki}$. Now what happens in the 2nd step of FW when we regress y_{it} on the vector of deviations from means? We get an OLS coefficient vector:

$$\left(\frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i)(x_{it} - \overline{x}_i) \right)^{-1} \frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i) y_{it}$$

A closer look at the within transformation

- But

$$
\begin{aligned}
\frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)y_{it} &= \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i+\overline{y}_i) \\
&= \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i)+\frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)\overline{y}_i \\
&= \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i) \\
&\quad +\frac{1}{N}\sum_{i}\overline{y}_i\underbrace{\frac{1}{T}\sum_{t}(x_{it}-\overline{x}_i)}_{=0} \\
&= \frac{1}{NT}\sum_{i,t}(x_{it}-\overline{x}_i)(y_{it}-\overline{y}_i)
\end{aligned}
$$

So we get the within estimator. This is also called a **"fixed effects"** estimator (since we add "effects" for each group).

Example: *Return to education using twins.*

- As we've discussed before, if you are looking at the relationship between wages and education, you might be concerned that there is an "omitted ability effect". Twins can help. Here our groups are "families" and we have 2 observations per family: twin 1 and twin 2. We assume:

  $$y_{fi} = \beta_1 + \beta_2 S_{fi} + \alpha_f + \epsilon_{fi} \qquad i = 1, 2$$

  where y = log wage, S = schooling. We can compare the conventional OLS model (stacking up the data for each twin) to a "within family" estimator.
- The data are from the "Princeton Twins Survey", a survey collected by Orley Ashenfelter and colleagues at the "Twinsburg Twins Festival" in the early 1990s. Researchers interviewed sets of twins over multiple years. (Ashenfelter and Rouse, QJE, 1998). For today we'll work with a set of 1416 twins (from 708 families).
- Here are the key results:

Example: *Return to education using twins.*

- OLS: log wage $= 1.310 + .0823$*Education $\quad R^2 = 0.084$;
  $MSE = 0.552$ standard error on education coefficient $= 0.007$
- Within: log wage $= 0.012 + .0683$*Education $\quad R^2 = 0.037$;
  $MSE = 0.521$ standard error on education coefficient $= 0.013$

- We can see that $\hat{\beta} = 0.082$ (std err=0.007) while $\hat{\beta}^W = 0.068$ (std err=0.013). The "within" estimator is a little smaller.

Control for α_i

- There is another way to think about the same problem. Assume:

$$y_{it} = x_{it}'\beta + \alpha_i + \epsilon_{it} \tag{8}$$

  We want to "control" for α_i. We could do this by introducing a full set of dummies for each person (which leads to the "within estimator", as shown in Lecture 8).

- A more sophisticated approach is to realize that if (8) is correct, then x_{is} does not directly affect y_{it} (for $s \neq t$) – but it may be useful in controlling for α_i.
- For example, in the $T = 2$ case we have $\{x_{i1}, x_{i2}\}$. We could consider regressing y_{i1} on both x_{i1} and x_{i2}. What will this do?

Control for α_i

- Let's assume $E[\epsilon_{is}|x_{it}] = 0$. (**strict exogeneity**, stronger than just the contemporary ϵ_{it} and x_{it} being uncorrelated). Then:

$$E[y_{i1}|x_{i1}, x_{i2}] = x_{i1}'\beta + E[\alpha_i|x_{i1}, x_{i2}] \tag{9}$$

- In the $T = 2$ case, suppose we believe that x_{i1}, x_{i2} are "equally informative" about α_i. This seems plausible in the twins case. Then:

$$E[\alpha_i|x_{i1}, x_{i2}] = \overline{x}_i'\gamma$$

- Assuming that α_i depends on the *average* of the $x's$ makes sense if we think there is nothing "special" about any particular x_{it}. In this case

$$\alpha_i = \overline{x}_i'\gamma + \phi_i \tag{10}$$

where ϕ_i has the property that

$$\forall t: \ E[\phi_i|x_{it}] = 0.$$

Control for α_i

- Combining 2 equations we get a different *estimating model*:

$$\begin{aligned} y_{it} &= x_{it}'\beta + \alpha_i + \epsilon_{it} \\ \alpha_i &= \overline{x}_i'\gamma + \phi_i \\ \Rightarrow y_{it} &= x_{it}'\beta + \overline{x}_i'\gamma + \phi_i + \epsilon_{it} \end{aligned}$$

  where now the remaining error is $\phi_i + \epsilon_{it}$ and $E[x_i(\phi_i + \epsilon_{it})] = 0$.
- So we can do OLS, using $\overline{x}_i$ as an extra set of regressors – $\overline{x}_i'$ to control for α_i.
- Note that if some element x_{kit} does not vary with time then $x_{kit} \equiv \overline{x}_{ki}$ – this regressor appears twice: once as part of x_{it}, once as part of $\overline{x}_i$. So this control function approach *won't work* for the non-varying $x's$ (equivalent to running a problematic multicolinear OLS).

Estimation: OLS with Control Function

- We have a new model to estimate by OLS:

$$y_{it} \quad = \quad x_{it}'\beta + \overline{x}_i'\gamma + \epsilon_{it}', \qquad \epsilon_{it}' = \phi_i + \epsilon_{it}$$

- What happens when we include $\overline{x}_i$ as an extra regressor? We need to use FW. We'll prove the case where $x_{it} = (1, z_{it})$ (one regressor of interest plus a constant). In this case $\overline{x}_i = \overline{z}_i$ (mean of constant is constant).
- From FW, we know

$$\hat{\beta}_2 = \left(\frac{1}{NT} \sum_{i,t} \hat{\xi}_{it}^2 \right)^{-1} \frac{1}{NT} \sum_{i,t} \hat{\xi}_{it} y_{it}$$

where $\hat{\xi}_{it}$ is the estimated residual after we regress z_{it} on $\overline{z}_i$ and constant.

$$\xi_{it} = z_{it} - E^*[z_{it}|1, \bar{z}_i]$$

Estimation: OLS with Control Function

- What happens when we regress z_{it} on $\bar{z}_i$ and a constant? The "auxiliary regression" is:

$$z_{it} = \pi_1 + \pi_2 \bar{z}_i + \xi_{it}.$$

The FOC's (1 and $\bar{z}_i$ are orthogonal to the residuals):

$$\sum_i \sum_t (z_{it} - \pi_1 - \pi_2 \bar{z}_i)\bar{z}_i = 0$$
$$\sum_i \sum_t (z_{it} - \pi_1 - \pi_2 \bar{z}_i)1 = 0.$$

$\widehat{\pi}_1 = 0, \widehat{\pi}_2 = 1$ solve the FOC's above if:

$$\frac{1}{NT} \sum_{i,t} (z_{it} - \bar{z}_i)\bar{z}_i = 0 \tag{11}$$

Verify (11) is true (exercise!).

Estimation: OLS with Control Function

So we end up with:

$$\hat{\beta}_2 = \left(\frac{1}{NT} \sum_{i,t} (z_{it} - \overline{z}_i)^2 \right)^{-1} \frac{1}{NT} \sum_{i,t} (z_{it} - \overline{z}_i) y_{it}$$

which is just the **"within"** estimator. The same algebra works when there are many rows in x_{it}.

3 ways to get within-estimator

We've shown that you can obtain the "within" estimator in 3 ways:

1. construct all the deviations from means and estimate

$$\hat{\beta}^W = \left(\frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i)(x_{it} - \overline{x}_i) \right)^{-1} \frac{1}{NT} \sum_{i,t} (x_{it} - \overline{x}_i)(y_{it} - \overline{y}_i)$$

2. estimate a model with dummies for each $i - group$:

$$y_{it} = x_{it}'\beta + (D_{1i}, D_{2i}, ...D_{Ni})\theta + \epsilon_{it}''$$

3. estimate a model that includes x_{it} and $\overline{x}_i$:

$$y_{it} \quad = \quad x_{it}'\beta + \overline{x}_i'\gamma + \epsilon_{it}'$$

The third approach is referred as control function $\overline{x}_i'\gamma$, or **correlated random effects (CRE)**.

Revisit Twins Study

- Let's take another look at the twins sample. Recall the model is

$$y_{fi} = \beta_1 + \beta_2 S_{fi} + \alpha_f + \epsilon_{fi} \qquad i = 1, 2$$

where y = log wage, S = schooling.

- Controlling for the mean $\bar{S}_f = \frac{S_{f1}+S_{f2}}{2}$: the estimating model is

$$y_{fi} = \beta_1 + \beta_2 S_{fi} + \gamma \overline{S}_f + \epsilon'_{fi}$$

What's the underlying assumption?

$$E[\alpha_f | S_{f1}, S_{f2}] = \gamma \overline{S}_f$$

.

- In the model for twin 1 we include his/her schooling and average schooling of twin 1 and twin 2 from the same family. We have a sample of 705 twin-pairs.

Revisit Twins Study

- Note that you can break apart the augmented model:

$$
\begin{aligned}
y_{fi} &= \beta_1 + \beta_2 S_{fi} + \gamma \overline{S}_f + \epsilon'_{fi} \\
&= \beta_1 + \beta_2 S_{fi} + \gamma[(S_{fi} + S_{f\sim i})/2] + \epsilon'_{fi} \\
&= \beta_1 + (\beta_2 + \gamma/2) S_{fi} + (\gamma/2) S_{f\sim i} + \epsilon'_{fi}
\end{aligned}
$$

  Which shows that we could just include S_{fi} and $S_{f\sim i}$, then get an estimate of $\gamma/2$ on sibling's education which we then subtract off the coefficient on own education to get a "clean" estimate of β.

- Sometimes people don't have information on sibling education so instead they use parent's education. Let's compare that estimate too.

Revisit Twins Study

Alternative Estimates of Return to Education Using Twins Sample

	OLS	OLS with Family Dummies	OLS with Mean Educ. Of Pair	OLS with Educ. Of Sibling	OLS with Educ. Of Father
Own Education	0.0802 (0.0071)	0.0678 (0.0131)	0.0678 (0.0197)	0.0749 (0.0106)	0.0843 (0.0075)
Mean Educ. of Pair	--	--	0.0143 (0.0212)	--	--
Sibling Education	--	--	--	0.0072 (0.0106)	--
Father's Education	--	--	--	--	-0.0097 (0.0052)